

Howard (Hao) Ye

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[LinkedIn](#) — [GitHub](#) — [Portfolio](#)

SUMMARY

Software engineer with experience across distributed systems, web applications, cloud infrastructure, and applied machine learning. Background includes production deployments, security hardening, data workflows, AI API integrations, and interactive systems built for healthcare, energy, and analytics use cases.

SKILLS

Backend & APIs: Node.js, REST APIs, Express, Go ([gorilla/mux](#)), Python, FastAPI, TypeScript

Frontend: Vue.js, React, JavaScript, HTML/CSS, Socket.IO

Data & ML: SQL, PySpark, PostgreSQL, SQLite, Redis, MLflow, Scikit-learn, Data Vault 2.0

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Jenkins, AWS, GCP, Ansible, Terraform

Security & Observability: JWT, API keys, rate limiting, TLS, Prometheus, Grafana, fail2ban, WireGuard
AI & LLM APIs: Google Gemini API, FastAPI, context injection pipelines, [@atproto/api](#), prompt engineering

Languages: Python, TypeScript, JavaScript, Go, Bash, Java, C/C++

EDUCATION

MS Computer Engineering (Research Track, formerly PhD candidate) — University of Tennessee Knoxville
2016–2025

EXPERIENCE

Graduate Research Assistant — **University of Tennessee Knoxville** **2017–2025**

Research: Performance Modeling of HPC Storage Systems and Federated Learning for Smart-Home Energy Optimization

- Collected and processed IOR benchmark output across sequential, random, strided, and shared-file workload patterns on a production Lustre HPC cluster to extract hyperparameters used to parameterize two independent performance models.
- Built a Colored Petri Net model in CPN Tools and a component-level discrete-event simulator in OMNeT++ (C++), then compared throughput, server utilization, and sojourn time predictions across both models; familiar with batch job scheduling patterns common in HPC environments.
- Identified data quality limitations in real Lustre OSS/OST telemetry, where incomplete and noisy server-side logs made ground-truth validation difficult and limited scaling to larger cluster configurations.
- Designed a federated learning system for smart-home energy optimization where each home trains locally on appliance usage data and outputs appliance schedules that minimize cost under grid pricing, battery storage, and solar generation constraints.

AI Software Engineer — **MatchingDonors.com (Remote)** **Jan 2026–Present**

- Rebuilt the public-facing matchingdonors.com/life experience to help patients and donors find relevant resources faster and more safely through AI-assisted interactions.
- Designed frontend and backend flows from scratch, including voice/text interaction by tab and role-specific navigation for the site refresh.
- Kept a simplified live version online while the larger admin branch remains under review by the team.

PROJECTS

Insurance Fraud Detection Platform on Microsoft Fabric — [\[Repo\]](#)

- Built an insurance claims fraud analytics platform to turn raw claim data into fraud-risk signals for investigation and reporting.
- Used Microsoft Fabric Lakehouse, PySpark, MLflow, and SQL analytics endpoint workflows to ingest, clean, engineer features, train models, and validate results on 15,420 claims.
- Trained and evaluated Decision Tree, Random Forest, and Gradient Boosted Tree classifiers; Random Forest reached AUC 0.94 and F1 0.87 on held-out data.

- Organized the pipeline into Bronze/Silver/Gold layers so raw data, curated features, and scored outputs could be reused by downstream analytics.

Insurance Claims Data Vault — [\[Repo\]](#)

- Built an end-to-end Microsoft Fabric pipeline to model multi-source insurance claims data into a governed, warehouse-ready format using Data Vault 2.0.
- Designed hubs, links, and satellites for core entities (claims, policies, customers, vehicles, payments) and orchestrated five notebook stages: Bronze ingest, Silver standardize, Raw Vault load, Gold mart build, and Warehouse publish.
- Delivered three business-facing SQL warehouse tables validated end-to-end in Microsoft Fabric: `wh_claim_360`, `wh_claim_history_timeline`, and `wh_fraud_support_case`.

Smart Home Energy Management System — [\[Repo\]](#) — [\[Demo\]](#)

- Built a smart-home energy management product to help users understand and optimize how solar, battery, grid, and appliances interact in real time.
- Implemented a Vue.js frontend with live charts and controls, a Node.js/TypeScript/Socket.IO simulation backend, a Python PPO reinforcement learning service, and a Go device-service microservice (`gorilla/mux`) exposing REST endpoints for device state.
- Modeled 11 appliance types including HVAC, water heater, and EV charging, with time-of-use pricing and thermal behavior to drive scheduling decisions.
- Orchestrated the system with Docker Compose and exposed a demo that illustrates the simulator and optimization loop end to end.

Server Automation, Hardening & Observability — [\[Repo\]](#)

- Built an 8-role Ansible project and supporting Terraform configuration to automate production Linux server provisioning, security hardening, and cloud resource creation for a VPN-oriented deployment.
- Added SSH key-only access, firewall default-deny, fail2ban, TLS renewal, Nginx reverse proxy, WireGuard VPN, and monitoring with Prometheus and Grafana; implemented blue-green deployment strategy to enable zero-downtime rollouts.
- Reduced server setup from hours of manual work to a repeatable run that completes in about 10 minutes.

Production REST API — [\[Repo\]](#) — [\[Live Demo\]](#)

- Built a live API to help doctors validate patient data quality with AI support and operational safeguards.
- Added rate limiting, API key authentication, health monitoring, and LLM response caching; covered auth, rate-limit enforcement, cache behavior, and error paths with an automated endpoint test suite.
- Deployed the service on Render so the system is publicly accessible as a working demo.